



Bridging Gaps in IoT Literature: A Thematic Categorization of Challenges and Resolutions

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Abstract. The Internet of Things (IoT) connects physical objects through networks, enabling autonomous operation and real-time data exchange across sectors such as healthcare, agriculture, smart cities, and industrial automation. As a rapidly evolving Complex Adaptive System, IoT faces major challenges in security, interoperability, data processing, and scalability. This literature survey systematically reviews the challenges and evaluates their proposed solutions such as Artificial Intelligence-based (AI-based) anomaly detection, predictive analytics, blockchain technologies, and Big Data frameworks like Hadoop, Spark, and Kafka. The paper also highlights standardized communication protocols and network optimization techniques in improving system efficiency. To illustrate strengths and limitations, reviewed techniques and methods are comparatively analysed and presented via graphical charts. In spite of advanced solutions, existing IoT architectures are required to be more secure, scalable and energy-efficient as some technical gaps still remain. Future work must focus on enhancing real-time data processing, ensuring scalability, and developing sustainable designs to support the widespread and reliable deployment of next-generation IoT systems.

Keywords: Internet of Things (IoT) · Security · Software-Defined Networking (SDN) · Federated Learning (FL) · Cryptography · Blockchain · Big Data

1 Introduction

Internet of Things (IoT) is a holistic platform that consists of devices embedded with sensors, processors, and communication technologies. Due to technological advancements, it has facilitated integrity and connectivity in day-to-day devices, thus increasing efficiency across a vast range of fields [14]. Healthcare systems in today's world need systematic and strategic integration of IoT technologies to address daily challenges effectively like remote patient monitoring, timely diagnosis, and resource optimization [3]. IoT operations involve a broad range of communication protocols that have the capability of handling large data flows from different devices. Without such protocols, interoperability and smooth operation of IoT devices in different applications would be impossible. While numerous protocols such as IPv6 over Low-power Wireless Personal Area Networks (6LoWPAN), Zigbee, Bluetooth Low Energy (BLE), Data Distribution

Service (DDS), Open Platform Communications Unified Architecture (OPC UA), Routing Protocol for Low Power and Lossy Networks (RPL), as well as Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) exist, MQTT and CoAP are among the predominantly used. MQTT is suitable for continuous data streaming, whereas CoAP is configured for lightweight, optimized transactions of documents and resources [14]. Such seamless interoperability requires managing IoT-generated data which is driven by technologies like cloud computing and big data infrastructure. Cloud-based platforms offer the scalability required for storage, analysis, and improved system performance. Even with these hopeful advancements, limitations on the underlying resources and diversity of IoT devices like limiting computation power, memory, and energy are major concerns for security. These limitations have the potential to increase cyber risks and botnet attacks based on IoT that result in disruption of entire networks [5]. Although communication efficiency and operations have become better with IoT, security and confidentiality issues continue to be addressed to ensure its long-term growth, build trust among users, and promote wider deployment [7]. Its further performance depends upon network connectivity and is restricted due to power consumption and resource limitations; never mind the diversity in their size and complexity. Furthermore, these devices need affordable and low-energy-consuming designs in order to facilitate smooth and continuous data sharing over various environments [9]. Deep Learning (DL) is among those advanced techniques that counter security threats. It analyzes the behavior patterns of device and sensor data to distinguish between ‘normal’ and ‘abnormal’ activities [12].

IoT technology has created such a group of interrelated smart devices that enables ongoing communication and real-time monitoring of devices with their surroundings, while remote control is enabled. This integration is driving the interaction of virtual space with the physical world, promoting innovation in fields such as smart cities, industrial automation, transportation, and home automation [29].

The structure of the paper is as follows: Sect. 2 contains an overall review of the literature. Section 4 determines the most significant challenges in the IoT environment. Section 5 discusses recent solutions and suggested techniques. Section 7 presents the conclusions of the main findings. Lastly, Sect. 8 presents prospective directions for future work.

2 Literature Review

The growth of IoT technology has led to better automation, higher efficiency, and smarter decision-making. However, a major limitation is that many IoT devices depend on batteries with limited capacities. This makes it important to optimize power use to prolong their lifespan. To resolve this, energy-saving methods such as advanced energy harvesting, low-power communication protocols, and intelligent energy management powered by AI have become increasingly important [6]. Furthermore, processing data near its point of origin through edge computing helps decrease the dependence on remote cloud servers, which in turn reduces overall energy usage and latency [11]. Security challenges remain a major concern, vulnerability of IoT devices towards cyberattacks, unauthorized access, and data breaches. Hence, ensuring strong encryption, secure authentication, and timely firmware updates is critical to safeguarding these systems [3]. Extensive data exchange

often leads to privacy challenges, thereby necessitating approaches like differential privacy and blockchain-driven authentication to strengthen data security [39]. Absence of standardized protocols contributes to fragmented IoT ecosystems, complicating seamless integration and cross-device communication between different manufacturers is challenging [17]. Middleware and open-source are proposed solutions to improve interoperability across diverse IoT platforms [8]. Challenges such as network congestion, latency, and bandwidth limitations increase as IoT networks scale. Technologies including Fifth Generation (5G), software-defined networking, and fog computing provide promising avenues to enhance network performance and scalability [22]. Partnerships that include researchers, industry experts, and policy-makers are essential for developing secure, efficient, and interoperable IoT systems for the future [24]. As urbanization increases, IoT has become a key factor in shaping smart city infrastructure. Using fog computing, real-time analytics, and information-centric networking is vital for managing the complex needs of urban areas while addressing ongoing issues like privacy, data protection, and resource limitations [4].

Even though IoT has the potential to transform fields like healthcare, agriculture, and smart homes, the variety of IoT devices and their limited resources create significant risks, especially the rising threat of IoT-based botnet attacks. While these attacks have received attention in recent years, thorough reviews and detailed strategies to tackle them are still scarce [5]. The transition from basic machine-to-machine communication to intelligent IoT systems has caused widespread changes in various industries. Nevertheless, the absence of unified architectural standards, combined with persistent issues related to interoperability, scalability, and security, continues to shape ongoing research efforts [6].

While innovation has progressed rapidly, issues regarding data privacy, system scalability, and security continue to dominate academic discussions and require thorough, interdisciplinary approaches. More studies have highlighted both the promise and challenges of IoT across various sectors [7].

At the same time, research into disruptive technologies like IoT, AI, and blockchain has shown that existing frameworks often struggle to define disruption clearly or provide useful guidance for early integration and long-term cooperation within organizations [10].

In industrial applications, combining AI with IoT has introduced powerful tools for predictive maintenance, operational insights, and process automation. However, this progress has increased the need for techniques that protect sensitive industrial and personal information. Adopting privacy-preserving techniques such as federated learning, secure multi-party computation, and differential privacy is increasingly recognized as essential for ensuring data security while keeping analytical value and meeting regulatory requirements [11].

3 Methodology and Analysis

The study conducts a Systematic Literature Review (SLR) to explore the progression, limitations, and innovations in the Internet of Things (IoT) ecosystem. The aim is to assess how communication protocols, architectural designs, and data processing methods

have evolved, along with the challenges that still hinder smooth implementation. A focused search was carried out across IEEE Xplore, SpringerLink, ScienceDirect, and Google Scholar. This search included literature published from 2013 to 2024. This timeframe was selected to showcase the major advancements in IoT technologies over the last decade. The keywords used were “IoT protocols,” “MQTT and CoAP,” “cloud computing in IoT,” and “IoT security.” We included only papers in English that offered technical depth and relevance to IoT systems, while excluding outdated or non-technical works.

The selected studies were categorized into three groups: communication frameworks, application-level challenges, and proposed solutions. We reviewed each paper to identify common themes and contributions. Each article was analyzed critically based on its goals, methods, contributions, and findings. We noted important factors such as protocol efficiency, scalability, latency, and security features when applicable.

We kept a data extraction sheet to maintain consistency in reviewing and synthesizing the chosen studies. The themes that emerged from this process were further checked through comparisons across papers to identify recurring trends, ongoing challenges, and research gaps.

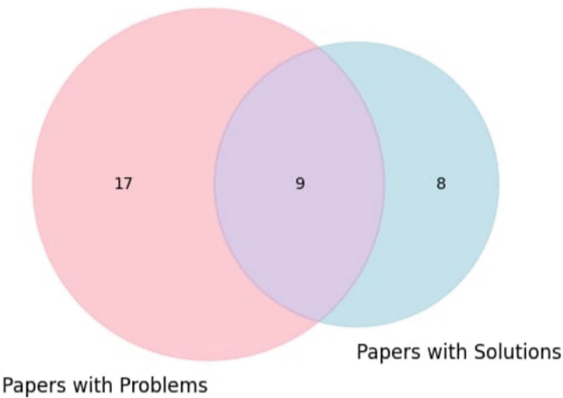


Fig. 1. Problem v/s Solution in Reviewed Papers

Out of 34 papers reviewed, 50% reveal gaps and limitations in current IoT systems while 23.5% highlight emerging solutions to these issues. 26.5% papers offered both shortcomings as well as their resolution. The Venn diagram in Fig 1 reveals more papers focused on identifying problems than on offering solutions (Fig. 2).

Among the collected papers from 2017 to 2024, approximately 35% of papers in 2018 and 2020 focused on Network and Communication, while 2021–2023 saw 30% on Security. Architecture and Big Data contributed less than 25% overall.

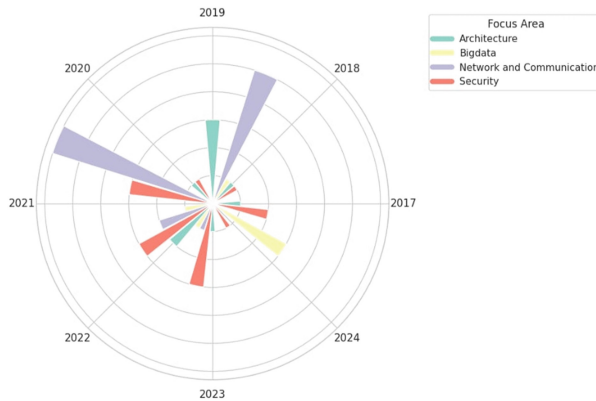


Fig. 2. IoT Research Focus by Area and Year

4 Highlighted Challenges in IoT

This section highlights current gaps and limitations in IoT practices.

4.1 Security and Privacy

Network security is at risk due to vulnerability of IoT systems to sophisticated malware, phishing, ransomware, and Distributed Denial of Service (DDoS) attacks [23]. Long-term system reliability is also reduced due to the use of outdated and counterfeit hardware, software, and network components [42]. Threat detection models that rely on Machine Learning (ML) and Deep Learning (DL) are vulnerable to adversarial inputs, leading to misclassifications [35]. Current defenses still largely lag behind evolving botnets despite advances in detecting Command and Control (C&C) servers and domain blacklisting [36]. Dynamic IoT environments require adaptive, interpretable, and highly resilient security solutions [27].

4.2 Interoperability

Interoperability is essential for integrating diverse systems but remains challenging due to heterogeneous hardware, software, and communication protocols. The pressure of rapidly evolving devices, fragmented frameworks, and inconsistent standards frequently limit real-time data exchange [21]. Centralized models face authentication issues as networks expand [27]. Due to strict proprietary designs, older protocols like Profibus and Modbus do not support real-time communication or interoperability [41]. More advanced and scalable frameworks are needed to enable consistent connectivity and communication [4, 38].

4.3 Network and Communication

IoT networks face growing challenges as connected devices and data demands surge. Many devices operate in bandwidth-limited environments and rely on wireless communication, making them prone to signal interference and noticeable delays [28]. Scalability

is severely hampered by the limited wireless spectrum and inefficient network-on-chip (NoC) communication [43]. Cloud-based AI adds latency, particularly in real-time applications like healthcare [32, 39]. Communication infrastructure is further impacted by network congestion, privacy concerns, and high deployment costs [29, 36].

4.4 Big Data

Integrating IoT with big data presents major challenges in managing vast structured and unstructured information. Efficient storage and processing are critically vital for real-time analytics and responsiveness [21, 26]. However, decision-making is frequently hampered by inconsistent data speed, availability, and trustworthiness [31]. Traditional cloud infrastructure introduces latency and bandwidth issues, severely limiting real-time use [36]. The convergence of various data sources further increases complexity and exposure risks [11]. Edge devices often lack the necessary capacity for advanced analytics, and scalability issues continue to obstruct actionable insights [2, 25, 34].

4.5 Architecture

IoT architectures often lack standardization, leading to fragmented and incompatible systems that fail to meet performance, security, and reliability demands [7, 30, 37]. It can be difficult to design flexible, resilient, and secure architectures, especially under cost constraints. Adoption is resource-intensive since implementation necessitates a thorough analysis of environments, user needs, and workflows [13]. Further, scalable deployment may be impeded by energy inefficiencies and architectural complexity, especially in battery-powered systems that require low-power solutions [1, 28, 33].

5 Proposed Solutions and Strategies

This section explores emerging approaches to resolve these obstacles.

5.1 Security and Privacy

To preserve privacy, emerging solutions allow computations on encrypted data through Homomorphic Encryption, enable secure collaboration on multiple devices using Secure MultiParty Computation (SMPC), support local training models that keep raw data private through Federated Learning (FL) [9, 11, 26]. IoT protocols like MQTT (Message Queuing Telemetry Transport) and CoAP (Constrained Application Protocol), Firewalls, Intrusion Detection Systems (IDS), Data Distribution Service (DDS), ZeroMQ provide network-level security [41]. Blockchain, owing to its decentralized nature, uses methods like group signatures, vector-based encryption, Multi-Access Edge Computing (MEC) and BlockBDM, enhancing secure data transmission among IoT devices [25]. To withstand evolving cyberattacks, algorithms such as Convolutional Neural Networks (CNNs), Stacked Autoencoders, and Extreme Learning Machines (ELMs) make use of machine learning concepts including Recurrent Neural Networks (RNNs) and Deep Neural Networks (DNNs) for effective anomaly detection [22].

5.2 Interoperability

For consistent interoperability, promising approaches include Software Defined Networking (SDN) as its design aligns with the concept of separation of concerns enabling abstraction and decoupling of fundamental tasks [20]. Mapping IoT protocols to existing smart grid standards facilitates live monitoring and control [41]. Reliability of data transmission is promised by deploying Quality of Service (QoS) mechanisms [15]. Unifying architectures are required to integrate various IoT devices; a Big IoT model that incorporates Tabdoc strategy has emerged as a valuable solution [40].

5.3 Network and Communication

IoT devices make use of popular protocols and technologies such as MQTT, CoAP, Wireless Sensor Networks (WSN), Low Range Wide Area Network Protocol (LoRaWAN), WiFi, Bluetooth, etc. Techniques like SigFox, which suit long-range transmission, and Worldwide Interoperability for Microwave Access (WiMAX) based on IEEE 802.16 standard are growing viable [19]. Cloud-based IoT networks' robustness is enhanced by Mixed Integer Linear Programming (MILP) [40]. A Software-Defined Wireless Sensor Network (SDWSN) model is put forward to address limited resources availability and diverse network configuration [24]. To improve communication among IoT devices connected with low bandwidth, the Side-chain approach of blockchain prevents the need for transmission of data with the entire system, optimizing performance [25].

5.4 Big Data

Big data technologies significantly improve the capacity of IoT devices to compute and process large volumes of data collected from sensors. For example, Flume, Kafka, and Apache Storm are appropriate tools for live data ingestion that will help incorporate continuous data streams in IoT systems, and Apache Spark is employed to enhance processing time, enabling real-time decision-making [21, 40]. Leximancer adds dimensions to IoT analytics as it facilitates multi-source data analysis and provides qualitative output from machine learning [10]. Big data infrastructures such as Spark and Hive can further augment IoT applications through predictive and statistical analyses of sensor-generated data. Furthermore, Artificial Intelligence and Edge Computing services on IoT platforms such as Smart Environmental Monitoring and Analytical in Real-Time (SEMAR) utilize the advantages of big data techniques for synchronous batch computation, for more efficient bandwidth usage and improved system speed and agility [36].

5.5 Architecture

The core foundation of IoT progress revolves around its architecture, the standardization of which is considered beneficial. Modern approaches such as Distributed Internet-like Architecture of Things (DIAT), DRAW model, Sensing and Actuation as a Service (SAaaS), and Microservices architectures have surfaced, along with frameworks like European Telecommunications Standards Institute (ETSI), Machine-to-Machine (M2M), and oneM2M [20, 40]. Clustering, scheduling algorithms, Dynamic Voltage

and Frequency Scaling (DVFS), Ultra-Low Power (ULP), Very-Large-Scale Integration (VLSI) circuits are techniques that have emerged to implement energy-efficient architecture [13, 18].

Since most edge IoT devices are resource-constrained, optimizing both computational load and energy consumption has become vital. Lightweight cryptography, model compression, and edge offloading strategies [24] can further improve energy efficiency and maintain security without overwhelming device capabilities. Integrating these mechanisms supports scalable, low-power IoT deployments capable of sustaining real-time operations within hardware and energy limits. Frameworks such as the Californium (Cf) CoAP, based on Internet Engineering Task Force (IETF) recommendations, facilitate heterogeneous IoT device communication [28].

6 Applications of IoT

6.1 Healthcare

IoT technologies have transformed healthcare through remote patient monitoring, early disease detection, and automated data collection, improving treatment outcomes and reducing costs [1, 3]. Wearable devices and the Internet of Medical Things (IoMT) enable continuous health tracking, emergency alerts, and medication management, while integration with AI and deep learning supports smarter diagnosis and decision-making. In smart cities, Intelligent Healthcare Systems (IHS) use medical sensors and wearable devices to provide timely monitoring, forward data for diagnosis, and enable early detection of infectious or chronic illnesses [3]. Advanced IoT-based healthcare applications combine thermal cameras, embedded sensors, and e-Health systems to share patient data with healthcare personnel, monitor vital signs, and track environmental conditions [25]. Examples include Electrocardiogram (ECG)-based arrhythmia detection using k-nearest neighbor algorithms, Global Positioning System (GPS)-enabled monitoring for patients with Alzheimer's or dementia, and wearable devices for soldiers' health tracking. While IoT offers significant potential, its large-scale adoption in healthcare remains in early stages, requiring further research to enhance health monitoring and management systems [31].

6.2 Agriculture

In the agriculture sector, IoT applications cover harvest monitoring, machinery management, water quality regulation, soil analysis, climate tracking, plant pathology and pest detection, precision farming, and greenhouse automation [15]. In crop farming, IoT-enabled sensors collect data on rainfall, leaf wetness, temperature, humidity, soil moisture, salinity, solar radiation, and pest movement. This information supports evidence-based management to boost harvest quality, reduce risks, and optimize revenue. For example, data from solar power sensors helps determine adequate sunlight exposure, while soil moisture readings assist in controlling soil conditions and preventing plant diseases. Accurate weather forecasts and pest movement tracking further enable preventive actions, improve planning, and reduce labor costs. Real-time data acquisition allows farmers to implement corrective measures promptly, enhancing both productivity and sustainability [15].

6.3 Smart Grid Environment/Energy

The future of traditional energy distribution lies in Smart Grids, combining it with renewable energy sources (RES), energy storage systems (ESS), and advanced control mechanisms to deliver stable, safe, and reliable electricity through bidirectional flow of both power and information [3, 41]. Legacy Supervisory Control and Data Acquisition (SCADA) systems did not have real-time capabilities, but IoT technologies like RFID, GPS, smart meters, and smart sensors provide live monitoring, fault detection, and adaptive control. IoT provides applications such as Advanced Metering Infrastructure (AMI), Demand Response (DR), and Home Energy Management Systems (HEMS) for dynamic load control, peak shaving, and ancillary services like frequency regulation. Electric vehicles (EVs) and microgrids act as prosumers through Vehicle-to-Grid (V2G) systems, providing grid stability and energy trading. Interoperability is the core of communication networks, including Home Area Networks (HAN), Field Area Networks (FAN), and Wide Area Networks (WAN), supporting secure and automatic energy management. With the integration of IoT devices within these areas, the smart grid gains self-healing capacity, improved reliability, and enhanced resilience to outages or cyberattacks [3, 41].

6.4 Supply Chain

IoT has drawn considerable focus with its ability to enhance accessibility, productivity, and strategic edge in business processes. IoT links distinctively identifiable products, machines, and people through sensors, storage and analytical platforms, and guidance systems that facilitate optimized solutions.

Industry projections confirm this exponential development, with Gartner estimating that by 2020 IoT would include some 26 billion interconnected devices, while Cisco put its potential market size at \$19 trillion. Leading organizations already use IoT to streamline supply chain operations. United Parcel Service (UPS), for example, embeds IoT technology into its fleet to cut idling time, decrease maintenance expenses, and increase operational safety. Freestyle Coca-Cola vending machines record customer-dictated beverage pairs, which are used to guide future product development. Disney's MagicBand optimizes guest experiences by minimizing wait times and maximizing satisfaction [10].

6.5 Environment Prediction and Protection

IoT technologies are increasingly applied to monitor and manage environmental conditions both indoors and outdoors. Indoor environmental IoT applications include smoke detection, fire alarms, light and motion sensing, and temperature and humidity monitoring whereas outdoor deployments encompass large scale applications such as traffic monitoring, storm activity detection, rainfall measurement, pollution tracking, and disaster detection for earthquakes, chemical hazards, and floods.

While indoor IoT systems generally offer higher reliability and lower maintenance costs due to their controlled environments, outdoor IoT deployments face challenges such

as harsh climate conditions, high maintenance requirements, and increased energy management demands. Addressing these challenges requires advanced solutions, including the integration of machine learning, blockchain, and industrial automation, to improve reliability and reduce operational costs. Addressing air pollution, one of the most significant environmental health risks can be achieved through real-time, fine-grained, and energy-efficient air quality monitoring systems using aerial and ground sensing as proposed in the studies [22].

In the broader context, IoT infrastructure when supported by technologies such as Narrowband IoT (NB-IoT) can serve environmental monitoring objectives while contributing to smart city development. Applications in smart transportation, intelligent buildings, and other public sectors leverage cloud computing and big data to optimize resource management and accelerate the creation of intelligent societies. Government involvement plays a crucial role in providing funding, creating incubation bases, promoting demonstration areas, and fostering Research and Development (R&D) and training initiatives. Successful deployment of such infrastructure requires coordinated planning between governments and telecom operators, as demonstrated by the Fuzhou municipal government's collaboration with Huawei and major operators to establish NB-IoT sample points, base stations, and an open laboratory [22].

7 Conclusion

The Internet of Things is changing numerous sectors by facilitating facile interconnection between disparate devices using sophisticated sensors, communication protocols, and data computation techniques. From smart homes and smart agriculture to industrial automation and healthcare monitoring, IoT solutions are streamlining processes, maximizing efficiency, and enhancing user experience in both urban and rural settings. Although live data management and prediction have been enhanced by protocols such as MQTT and CoAP as well as AI-based analytics, there are key challenges. These involve big data management challenges, connectivity, interoperability, scalability, and limitations in architecture. Such challenges do not only impact device performance but also come with the risks of system longevity and ongoing scalability, particularly as IoT environment becomes increasingly complicated and globally integrated. Further, the absence of standardization, safety loopholes, and limited resources decelerate efficient IoT deployment. Redressing these challenges necessitates solid and secure IoT frameworks that can efficiently handle voluminous data, ensure instant responsiveness, and secure data integrity and privacy. As IoT advances, priorities such as energy-efficient computing and ethical usage of data especially in ways ensuring equitable access and sustainable development need to be put into consideration as well. As IoT continues to grow in sectors such as smart cities, healthcare, agriculture, and manufacturing, perfecting its infrastructure will be crucial. Through ongoing innovation and a drive to overcome existing challenges, IoT can reshape the way information is collected, processed, and used to support intelligent, data-driven decision-making across the most important application areas. The way forward needs collective action by researchers, policy-making units, and industries to create trust, promote security, and maximize the utility of IoT technologies.

8 Future Scope

There is an absence of uniform standards which limits encoding/decoding of data, weakens new solutions by causing issues in interoperability, security, and architecture design. It would be beneficial to build global security frameworks that incorporate robust authentication, encryption, and risk evaluation. Integrating blockchain and federated learning with edge computing can enhance protection. Better public datasets and interpretable AI models are needed to overcome hindrance caused by scarcity of IoT botnet datasets. Protection against adversarial ML/DL attacks must upgrade, along with other security techniques like homomorphic encryption, differential privacy, and quantum-safe cryptography to secure low-resource devices. Since there is a vast inconsistency in data from different sources in IoT devices that makes computing more complex; new solutions should aim to transform raw data into normalized insights regardless of processing limitations. Simplifying and normalizing data inputs can improve algorithm performance and system scalability. Evolving solutions can work upon adopting middleware and AI-driven protocols to maintain consistent communication supporting a common framework. A well-known approach is Edge computing that can provide low-latency and real-time communication. Future solutions should be compatible with tech like 5G, Low Power Wide Area Network (LPWAN), and satellite IoT.

References

1. Abid, M.A., et al.: Evolution towards smart and software-defined Internet of Things. *AI* **3**(1), 100–123 (2022)
2. Adhicandra, I., Tanwir, T., Asfahani, A., Sitopu, J.W., Irawan, F.: Latest innovations in Internet of Things (IoT): digital transformation across industries. *Innov. J. Soc. Sci. Res.* **4**(3), 1027–1037 (2024)
3. Al-Rawashdeh, M., Keikhosrokiani, P., Belaton, B., Alawida, M., Zwiri, A.: IoT adoption and application for smart healthcare: a systematic review. *Sensors* **22**(14), 5377 (2022)
4. Al-Turjman, F., Zahmatkesh, H., Shahroze, R.: An overview of security and privacy in smart cities' IoT communications. *Trans. Emerg. Telecommun. Technol.* **33**(3), e3677 (2022)
5. Ali, I., et al.: Systematic literature review on IoT-based botnet attack. *IEEE Access* **8**, 212220–212232 (2020)
6. Alloui, H., Mourdi, Y.: Exploring the full potentials of IoT for better financial growth and stability: a comprehensive survey. *Sensors* **23**(19), 8015 (2023)
7. Alshohoumi, F., Sarrab, M., AlHamadani, A., Al-Abri, D.: Systematic review of existing IoT architectures security and privacy issues and concerns. *Int. J. Adv. Comput. Sci. Appl.* **10**(7) (2019)
8. Alsoufi, M.A., et al.: Anomaly-based intrusion detection systems in IoT using deep learning: a systematic literature review. *Appl. Sci.* **11**(18), 8383 (2021)
9. Ansari, D.B., Rehman, A.U., Ali, R.: Internet of Things (IoT) protocols: a brief exploration of MQTT and CoAP. *Int. J. Comput. Appl.* **179**(27), 9–14 (2018)
10. Aryal, A., Liao, Y., Nattuthurai, P., Li, B.: The emerging big data analytics and IoT in supply chain management: a systematic review. *Supply Chain Manag. Int. J.* **25**(2), 141–156 (2018)
11. Ayuns, L.: Privacy-preserving AI analytics for industrial IoT data: techniques and protection (2024)
12. Chen, J.I.Z., Lai, K.L.: Internet of Things (IoT) authentication and access control by hybrid deep learning method-a study. *J. Soft Comput. Paradig. (JSCP)* **2**(04), 236–245 (2020)

13. Dachyar, M., Zagloel, T.Y.M., Saragih, L.R.: Knowledge growth and development: Internet of Things (IoT) research, 2006–2018. *Heliyon* **5**(8) (2019)
14. Elhadi, S., Marzak, A., Sael, N., Merzouk, S.: Comparative study of IoT protocols. In: *Smart Application and Data Analysis for Smart Cities (SADASC 2018)* (2018)
15. Elijah, O., Rahman, T.A., Orikumhi, I., Leow, C.Y., Hindia, M.N.: An overview of Internet of Things (IoT) and data analytics in agriculture: benefits and challenges. *IEEE Internet Things J.* **5**(5), 3758–3773 (2018)
16. Fahim, M., Sillitti, A.: Anomaly detection, analysis and prediction techniques in IoT environment: a systematic literature review. *IEEE Access* **7**, 81664–81681 (2019)
17. Fahmideh, M., Zowghi, D.: An exploration of IoT platform development. *Inf. Syst.* **87**, 101409 (2020)
18. Farhan, L., Kharel, R., Kaiwartya, O., Quiroz-Castellanos, M., Alissa, A., Abdulsalam, M.: A concise review on Internet of Things (IoT)-problems, challenges and opportunities. In: *2018 11th International Symposium on Communication Systems, Networks & Digital Signal Processing (CSNDSP)*, pp. 1–6. IEEE, July 2018
19. Farooq, M.S., Riaz, S., Abid, A., Umer, T., Zikria, Y.B.: Role of IoT technology in agriculture: a systematic literature review. *Electronics* **9**(2), 319 (2020)
20. Gill, A.Q., Behbood, V., Ramadan-Jradi, R., Beydoun, G.: IoT architectural concerns: a systematic review. In: *Proceedings of the Second International Conference on Internet of things, Data and Cloud Computing*, pp. 1–9, March 2017
21. Hajjaji, Y., Boulila, W., Farah, I.R., Romdhani, I., Hussain, A.: Big data and IoT-based applications in smart environments: a systematic review. *Comput. Sci. Rev.* **39**, 100318 (2021)
22. Hou, J., Li, B.: The evolutionary game for collaborative innovation of the IoT industry under government leadership in China: an IoT infrastructure perspective. *Sustainability* **12**(9), 3648 (2020)
23. Hussain, F., Hussain, R., Hassan, S.A., Hossain, E.: Machine learning in IoT security: current solutions and future challenges. *IEEE Commun. Surv. Tutor.* **22**(3), 1686–1721 (2020)
24. Imran, Ghaffar, Z., Alshahrani, A., Fayaz, M., Alghamdi, A.M., Gwak, J.: A topical review on machine learning, software defined networking, Internet of Things applications: research limitations and challenges. *Electronics* **10**(8), 880 (2021)
25. Imran, M., Zaman, U., Imran, Imtiaz, J., Fayaz, M., Gwak, J.: Comprehensive survey of IoT, machine learning, and blockchain for health care applications: a topical assessment for pandemic preparedness, challenges, and solutions. *Electronics* **10**(20), 2501 (2021)
26. Jawad, A., Abu-AlShaeer, M.: Revolutionizing communication: how 5G, UAVs, and cloud technologies are shaping the new telecommunications landscape. *Pidvodni tehnologii* **1**(14), 75–84 (2024)
27. Karie, N.M., Sahri, N.M., Yang, W., Valli, C., Kebande, V.R.: A review of security standards and frameworks for IoT-based smart environments. *IEEE Access* **9**, 121975–121995 (2021)
28. Khanna, A., Kaur, S.: Internet of things (IoT), applications and challenges: a comprehensive review. *Wirel. Pers. Commun.* **114**, 1687–1762 (2020)
29. Kiran, M.B., Wynn, M.G.: The Internet of Things in the corporate environment: cross-industry perspectives and implementation issues. In: *Handbook of Research on Digital Transformation, Industry Use Cases, and the Impact of Disruptive Technologies*, pp. 132–148. IGI Global (2022)
30. Maroua, B., Abdelaziz, M.: Smart farming architectures based on IoT review: comparative study. *Procedia Comput. Sci.* **203**, 783–788 (2022)
31. Mohamad Jawad, H.H., Bin Hassan, Z., Zaidan, B.B., Mohammed Jawad, F.H., Mohamed Jawad, D.H., Alredany, W.H.D.: A systematic literature review of enabling IoT in healthcare: motivations, challenges, and recommendations. *Electronics* **11**(19), 3223 (2022)

32. Morillo, P., Orduña, J.M., Fernández, M., García-Pereira, I.: Comparison of WSN and IoT approaches for a real-time monitoring system of meal distribution trolleys: a case study. *Future Gener. Comput. Syst.* **87**, 242–250 (2018)
33. Motta, R.C., Silva, V., Travassos, G.H.: Towards a more in-depth understanding of the IoT paradigm and its challenges. *J. Softw. Eng. Res. Dev.* **7**, 3-1 (2019)
34. Nadella, G.S., Gonaygunta, H.: Enhancing cybersecurity with artificial intelligence: predictive techniques and challenges in the age of IoT. *Int. J. Sci. Eng. Appl.* **13**(04), 30–33 (2024)
35. Nazir, A., et al.: Advancing IoT security: a systematic review of machine learning approaches for the detection of IoT botnets. *J. King Saud Univ. Comput. Inf. Sci.* **35**(10), 101820 (2023)
36. Panduman, Y.Y.F., Funabiki, N., Fajrianti, E.D., Fang, S., Sukaridhoto, S.: A survey of AI techniques in IoT applications with use case investigations in the smart environmental monitoring and analytics in real-time IoT platform. *Information* **15**(3), 153 (2024)
37. Rozi, F.: Systematic Literature Review pada Analisis Prediktif dengan IoT: Tren Riset, Metode, dan Arsitektur. *Jurnal Sistem Cerdas* **3**(1), 43–53 (2020)
38. Shah, R., Chircu, A.: IoT and AI in healthcare: a systematic literature review. *Issues Inf. Syst.* **19**(3) (2018)
39. Sharma, V., Tripathi, A.K.: A systematic review of meta-heuristic algorithms in IoT based application. *Array* **14**, 100164 (2022)
40. Swamy, S.N., Kota, S.R.: An empirical study on system level aspects of Internet of Things (IoT). *IEEE Access* **8**, 188082–188134 (2020)
41. Tightiz, L., Yang, H.: A comprehensive review on IoT protocols' features in smart grid communication. *Energies* **13**(11), 2762 (2020)
42. Tweneboah-Koduah, S., Skouby, K.E., Tadayoni, R.: Cyber security threats to IoT applications and service domains. *Wirel. Pers. Commun.* **95**, 169–185 (2017)
43. Zikria, Y.B., Ali, R., Afzal, M.K., Kim, S.W.: Next-generation Internet of Things (IoT): opportunities, challenges, and solutions. *Sensors* **21**(4), 1174 (2021)